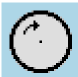
Image



Description

A typical example is in the package shipping industry where all packages have to be rotated to a uniform direction so that a scanner can automatically read the bar code holding the address information. The *Turnplate* has a capacity of one, i.e., only one *MU* can be located on it at any one time. The *MU* moves onto the *Turnplate* and the *Turnplate* starts rotating when the booking point of the *MU* has reached the center of rotation of the *Turnplate*. When the rotation is finished, the *MU* exits the *Turnplate*. The conveying direction on the *Turnplate* is unidirectional, i.e., the *MU* cannot be conveyed forward and then backward.

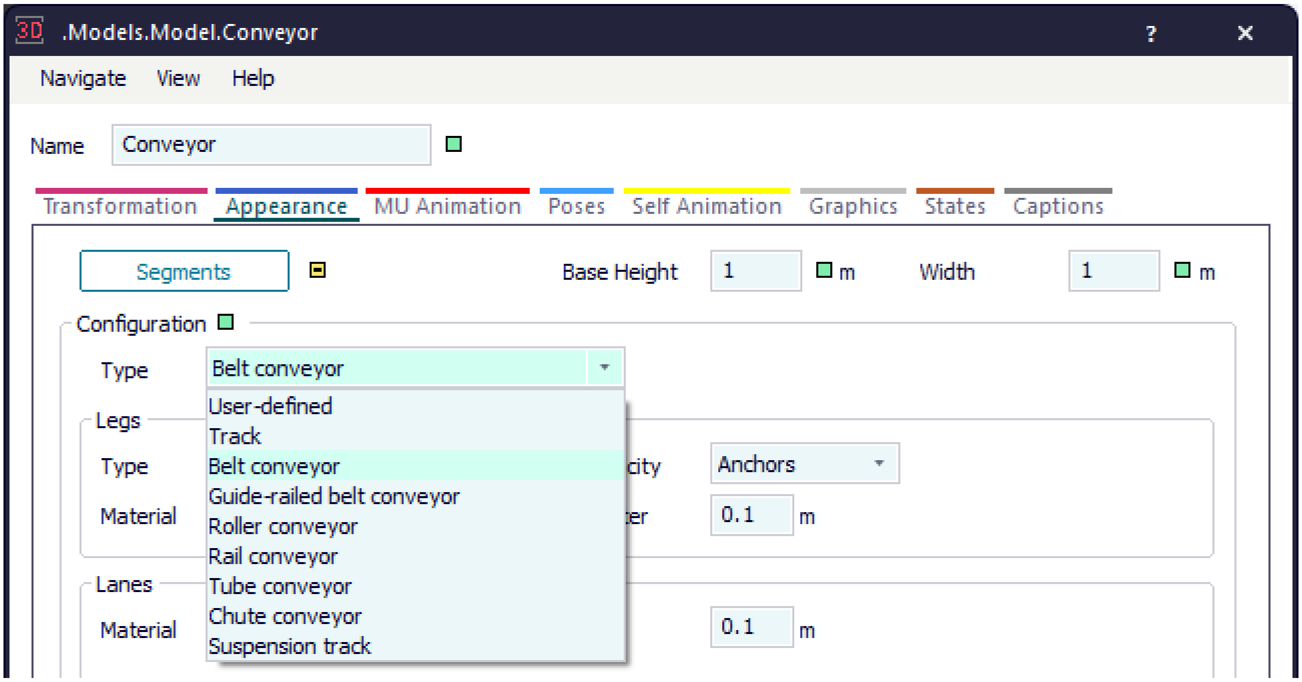
The length of the *MU* must not be longer than the **Length** of the *Turnplate* itself.

The center of rotation is by default located in the center of the *Turnplate*. The rotation takes up a certain amount of time. You can type in the rotation time per rotation steps of 90 degrees. If you type in a rotation time of 0, the *MU* is rotated instantaneous without using up any time at all.

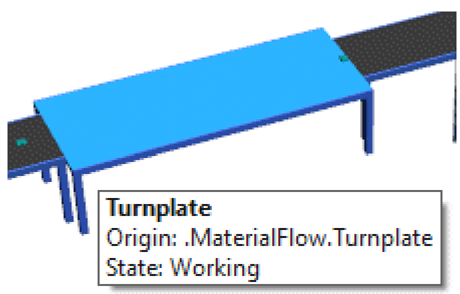
The rotation angle that you type in should be a multiple of 90 degrees. If you type in an angle other than that, *Plant Simulation* rounds this angle to the next angle that is divisible by 90. You can also type in a value greater than 360 degrees as long as it is divisible by 90. This way the *Turnplate* can rotate the *MU* several times to simulate packing machines, such as shrink wrappers, etc. By default, the *Turnplate* rotates 90 degrees clockwise.

After the turnplate has rotated the *MU* and it has left it, the *Turnplate* returns to its starting position.

You can select different configurations for the *Turnplate* on the tab **Appearance of the Length-oriented Objects**.



To show a **tooltip** with information about the *Turnplate*, hover with the mouse over it.

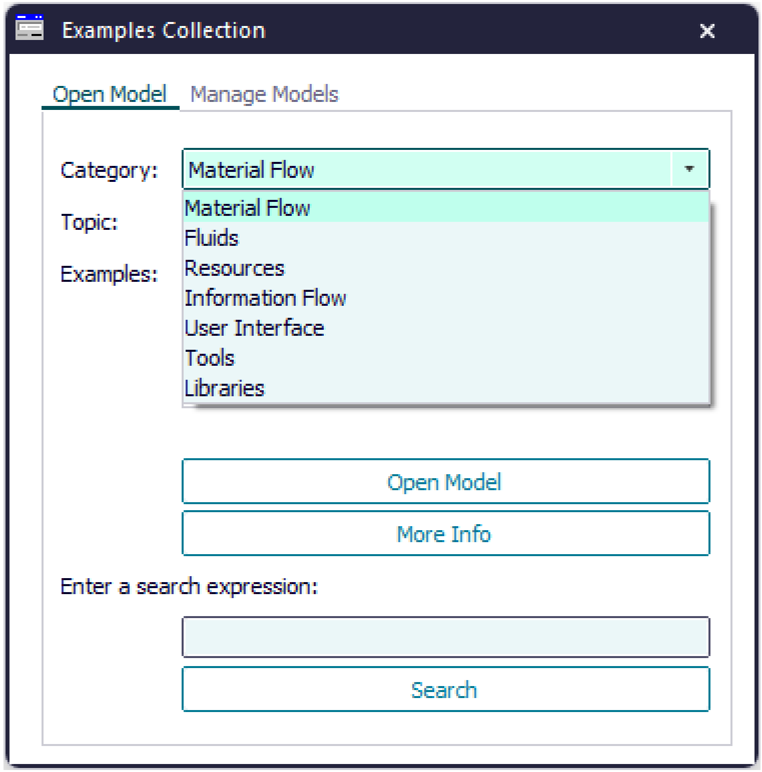


To change the length of the graphic and the anchor points of the *Turnplate*, click **Show Manipulators**  on the **Edit** ribbon tab or press **M** on the keyboard.

Add the Object to the Simulation Model

To add the object *Turnplate* to your simulation model, click  **Manage Class Library** > **Basic Objects > MaterialFlow > Turnplate** on the **Home** ribbon tab.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



See also

Align and Shrink-Wrap Parts with the Turnplate

Configure the Turnplate to Rotate the Part According to an Attribute

Video on YouTube

https://youtu.be/hOvdrDnvXXo?si=Cs5gOF4JB5PBlVma&t=532

Dialog Box of the Turnplate

Dialog Box of the Turnplate

Double-click the icon of the *Turnplate* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under **Dialog Items of the Objects**.

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box **Edit 3D Properties**:

- Click the button **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click **Show Manipulators** on the **Edit** ribbon tab or press **M** on the keyboard.

Tab Attributes

Tab Attributes

The tab **Attributes** provides the settings, which the object offers.

The settings are listed in the table of contents to the left.

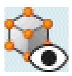
The shared properties are described under the **Tab Attributes**.

Length [text box] - Turnplate

Type the **Length** of the *Turnplate* into the text box.

Remarks

After you inserted it, *Plant Simulation* shows its length here. The *Turnplate* only rotates *MUs*, which it can accommodate in their entirety, meaning that they are shorter or as long as the value you enter here.

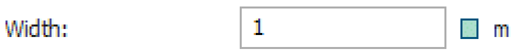
You can change the length of the graphic and the anchor points of the *Turnplate* by clicking **Show Manipulators**  on the **Edit** ribbon tab or by pressing **M** on the keyboard.

SimTalk

Length [Turnplate]]

Width [text box] - Turnplate

Type the **Width** of the *Turnplate* into the text box.



SimTalk

Width [SimTalk] - Turnplate

Speed [text box] - Turnplate

Type the **Speed** into the text box with which the *Turnplate* conveys the *MU* while it is located on the plate.

Remarks

You can also type in `-1` for an infinite speed.

SimTalk

Speed [SimTalk] - Turnplate

Rotation Time per 90° [text box] - Turnplate

Type the **Time** into the text box, which it takes the *Turnplate* to rotate by 90 degrees.

Remarks

To rotate the *MU* immediately, without using up any time at all, type in a rotation time of `0`.

SimTalk

RotationTimePer90Degrees [SimTalk] - Turnplate

Strategy [drop-down list] - Turnplate

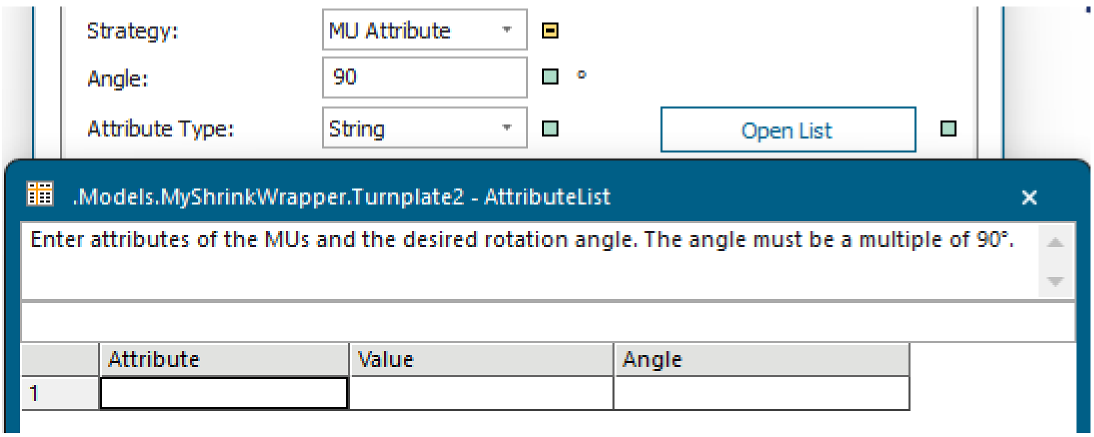
Select the **Strategy** according to which the *Turnplate* rotates the part.

Remarks

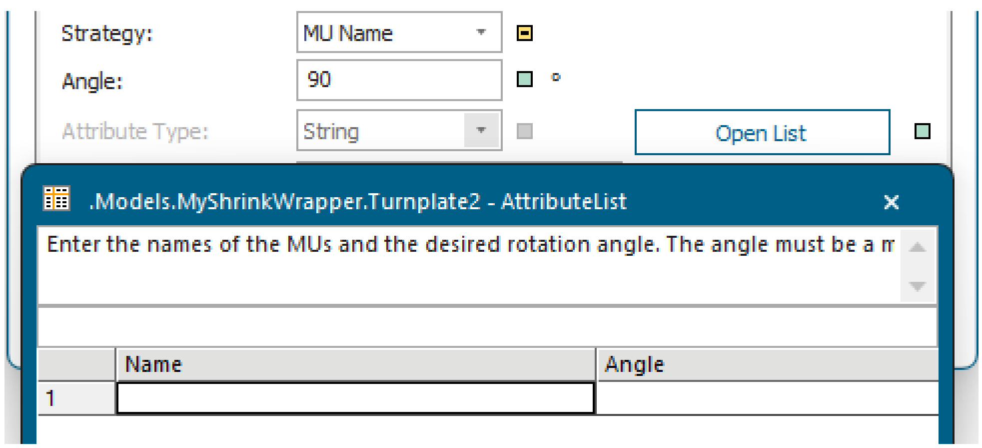
You can select one of these settings:

- **Angle** rotates the *MU* according to the rotation **Angle**, which you type in.

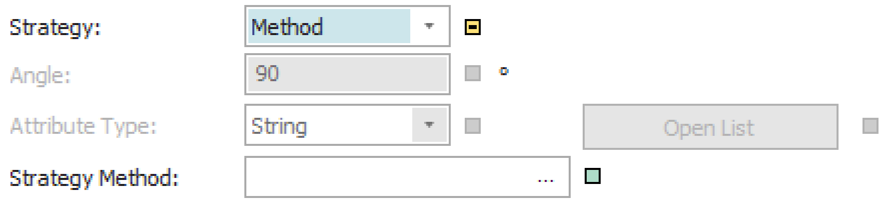
- **MU Attribute** rotates the *MU* according to a built-in or a *user-defined attribute* of the *MU*. Click **Open List** and type in the name of the **Attribute** of the *MU*, the **Value** of the attribute, and the rotation **Angle**.



- **MU Name** rotates the *MU* according to its name. Click **Open List** and type in the **Name** of the *MU* and the rotation **Angle**.



- **Method** rotates the *MU* according to the *strategy method*. Within this method you have to call the method **rotatePart** with the rotation angle as parameter. Type the name of the method into the text box **Strategy Method**.



SimTalk

Strategy [SimTalk] - Turnplate

See also

Configure the Turnplate to Rotate the Part According to an Attribute

Angle [text box] - Turnplate

Open List [Turnplate]

Strategy Method [Turnplate]

Angle [text box] - Turnplate

Type the **Angle** in degrees into the text box to which the *Turnplate* rotates the *MU*.

Remarks

The rotation angle should be a multiple of 90. If you type in an angle other than that, *Plant Simulation* rounds this angle to the next angle that is divisible by 90. You can also type in a value greater than 360 degrees as long as it is divisible by 90. This way the turnplate can rotate the part several times around its own axis to simulate packing machines.

By default, the turnplate rotates the part 90 degrees clockwise. To rotate the part counter-clockwise, type in negative angles.

SimTalk

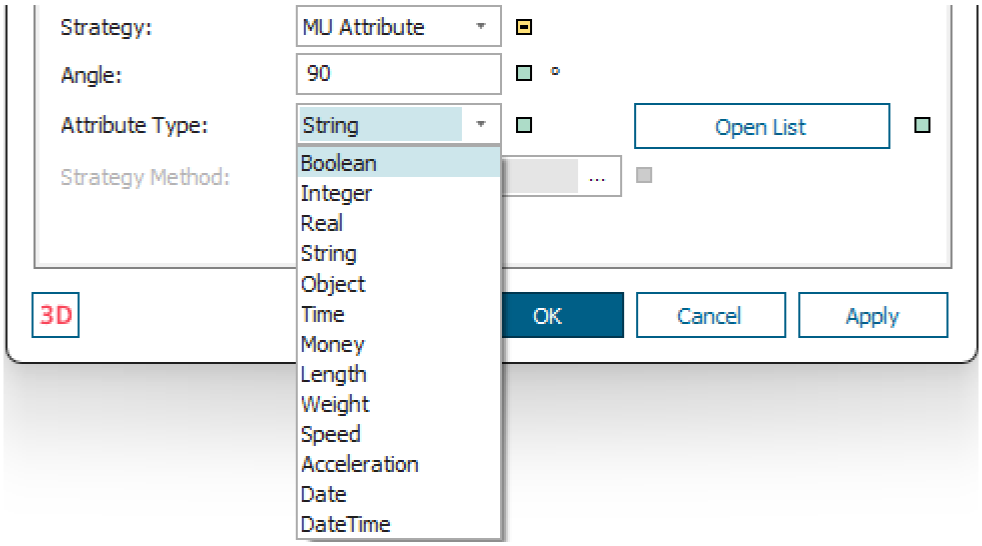
Angle [SimTalk] - Turnplate

Attribute Type [drop-down list] - Turnplate

Select the data type of the attribute that determines the rotation angle by which the *Turnplate* rotates the part.

Remarks

This applies to the **Strategy** > **MU Attribute**.



SimTalk

AttributeType [SimTalk] - Turnplate

See also

Strategy [drop-down list] - Turnplate

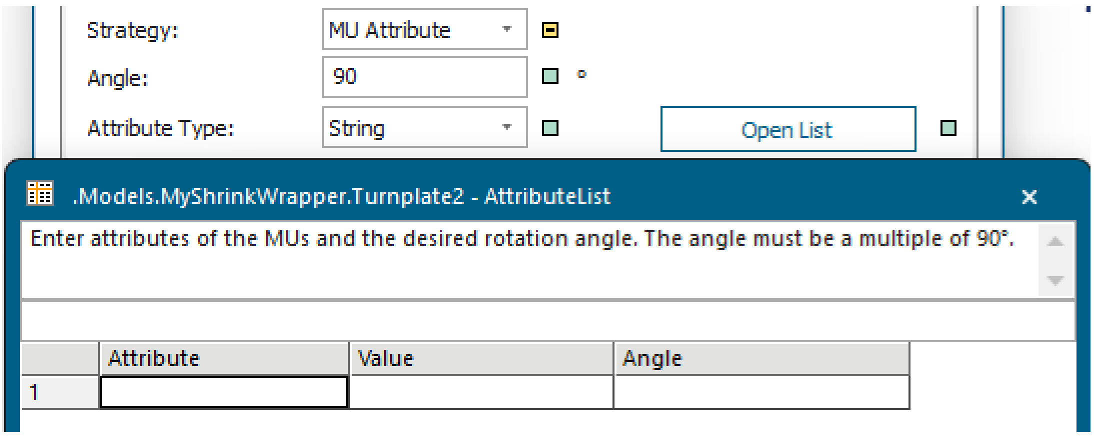
Open List [Turnplate]

To open the rotation angles list for the *MUs*, click this button.

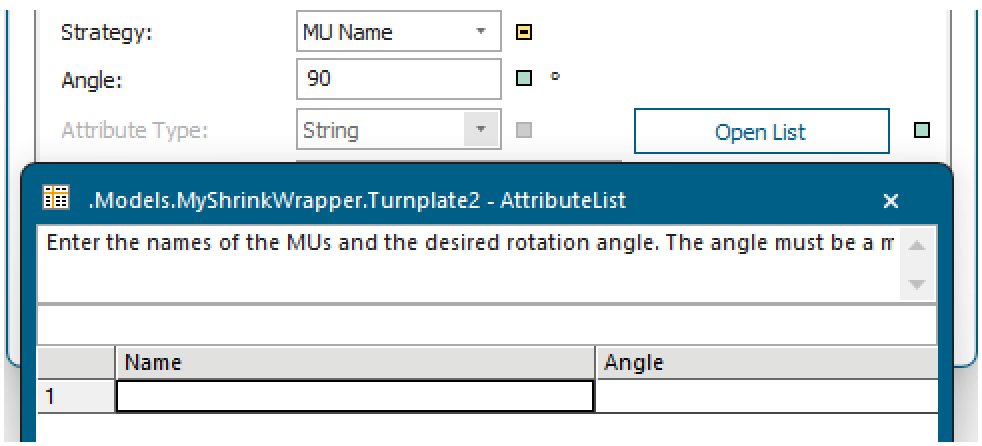
Remarks

The **Strategy** determines what you type in:

- For the **Strategy > MU Attribute** you can type in the name of the **Attribute** of *MU* part, the **Value** of the attribute, and the rotation **Angle**.



- For the **Strategy > MU Name** you can type in the **Name** of the attribute of the *MU* and the rotation **Angle**.



SimTalk

setAttributeList [SimTalk] - Turnplate

getAttributeList [SimTalk] - Turnplate

See also

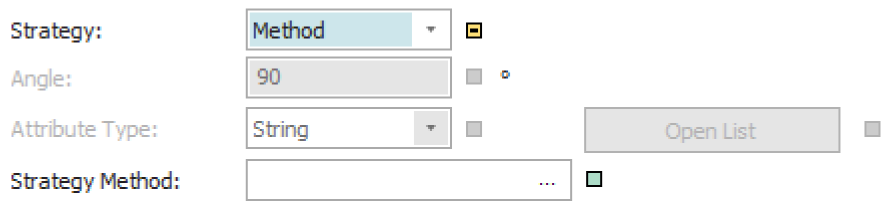
Strategy [drop-down list] - Turnplate

Configure the Turnplate to Rotate the Part According to an Attribute

Data Held in Tabular Form in Attributes [material flow objects]

Strategy Method [Turnplate]

Modifies the built-in behavior of the object. The object calls the **Strategy Method** as soon as the booking point of the part is located on the center of rotation of the *Turnplate*.



Remarks

The **Strategy Method** rotates the *MU* around the rotation angle which you specify.

Within the **Strategy Method** you have to call the method *rotatePart* with the rotation angle as parameter.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog **Select Object [for controls]** and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.

Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

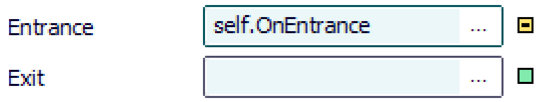
Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select **Create Control [context menu]**. *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.

- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.

- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

The **default strategy method** as a *user-defined attribute* looks like this:

```
?.rotatePart(90)
```

SimTalk

StrategyCtrl [SimTalk] - Turnplate

rotatePart [SimTalk]

See also

Select Object [for controls]

Configure the Turnplate

Strategy [drop-down list] - Turnplate

Automatic Stop [check box] - Turnplate

To set the **Current Speed** of the *Turnplate* to 0 when it does not transport a *MU*, select this check box.

Remarks

Automatically stopping the *Turnplate* might, for example, be the case if it is empty or if it is blocked because a *MU* cannot leave it.

If the **Speed** of the *Turnplate* is 0, the **Energy State** changes to **Operational**.

SimTalk

AutomaticStop [SimTalk] - Turnplate

Tab Times

Define **Times** as described under the **Tab Times**.

Select a distribution from the drop-down list and type the values that this distribution requires into the text box. *Plant Simulation* shows the parameters along the upper border of the tab. You can also select a constant time (**Const**).

You can set the type of the distribution and a complete set of parameters with the method **setTypeAndAttr [SimTalk]**.

See also

Recovery Time [general description]

Recovery Time Starts

Cycle Time [general description]

Tab Failures

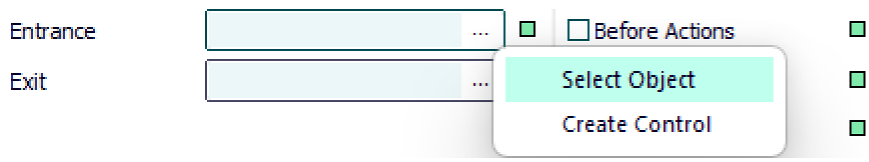
Define **failures** as described under the **Tab Failures**.

Tab Controls

Provides controls to modify the built-in behavior of the object.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog **Select Object [for controls]** and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



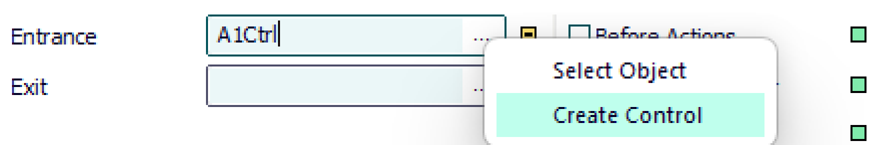
Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

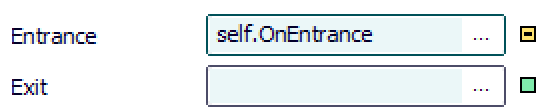
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select **Create Control [context menu]**. *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.

- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.

  Type the source code of this control into the *Method* that opens.

  To edit the source code later on:

  - Press **F2**.
  - Or hold down **Shift** and double-click into the text box.
  - Or select **Open Object** on the context menu.
  - Or click the tab **User-defined** and double-click the name of the *Method* in the list.

- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

See also

Entrance Control [general description]

Exit Control [general description]

Pull Control [general description]

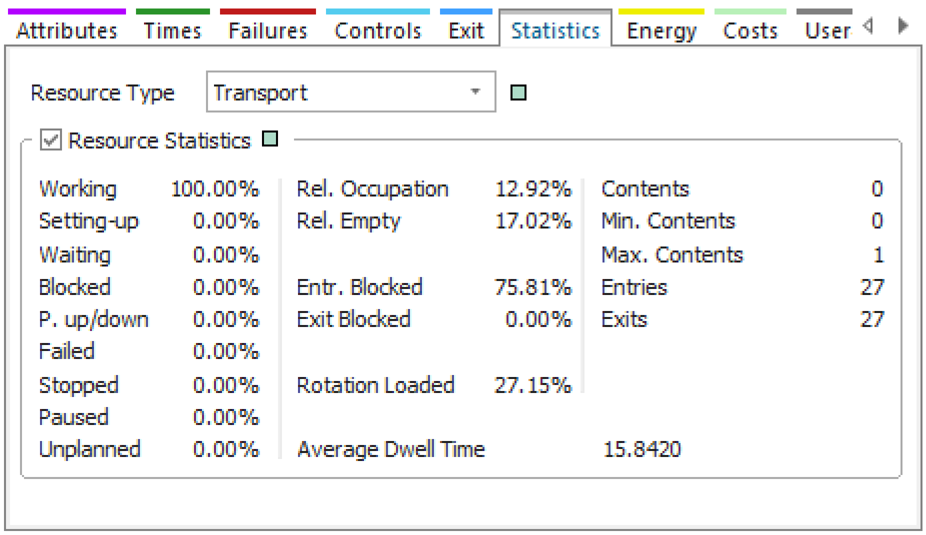
Shift Calendar [tab Controls]

Tab Statistics [Turnplate]

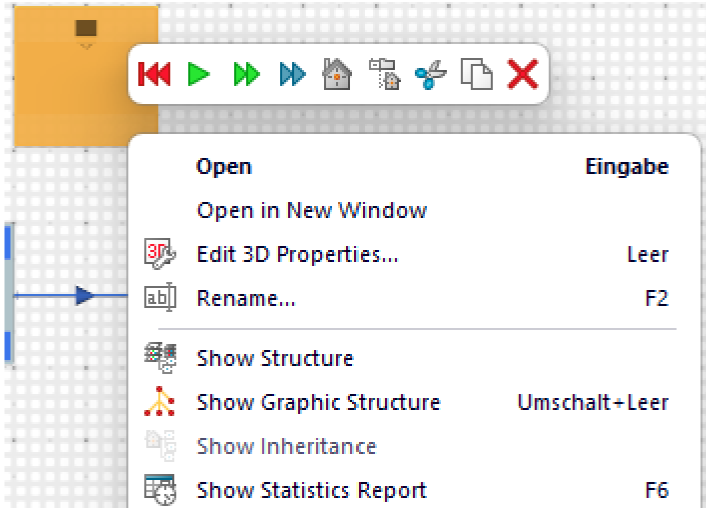
Statistics is described under the **Tab Statistics**.

In addition, the *Turnplate* collects this statistical value:

Item English	Description	Read-only attribute	Item German
Rotation Loaded	Shows the portion of the statistics collection period during which the *Turnplate* was rotating while transporting a *MU*.	**StatRotationLoadedPortion [SimTalk] - Turnplate**	**Drehung belegt**



To view the **Rotation Time** in the **Statistics Report**, select **View > Show Statistics Report** in the dialog of the object. You can also click the right mouse button in the *Frame* and select **Show Statistics Report** or you can press **F6**.



See also

Resource Statistics [check box]

Resource Type

Tab Energy

Select **energy settings** for the object on the **Tab Energy**.

Tab Costs

Select **costs settings** for the object on the **Tab Costs**.

While the *Turnplate* transports *MUs* costs accrue which result from the sum of the total investment costs and the total operating costs.

> **Note**
>
> The total investment costs only accrue during the **Depreciation Period**.

If the *Turnplate* is empty, the costs remain with the *Turnplate* as **general costs**.

See also

Simulate the Accrued Costs of the Machines

CostAnalyzer > **How the CostAnalyzer Assigns Costs to Part Types**

CostAnalyzer > **Costs Shown in the Costs Report**

Tab User-defined

Define your own attributes as described under the **Tab User-defined**.

Navigate Menu

The commands are described under the **Navigate Menu**.

View Menu

The **View Menu** provides commands to access its functions.

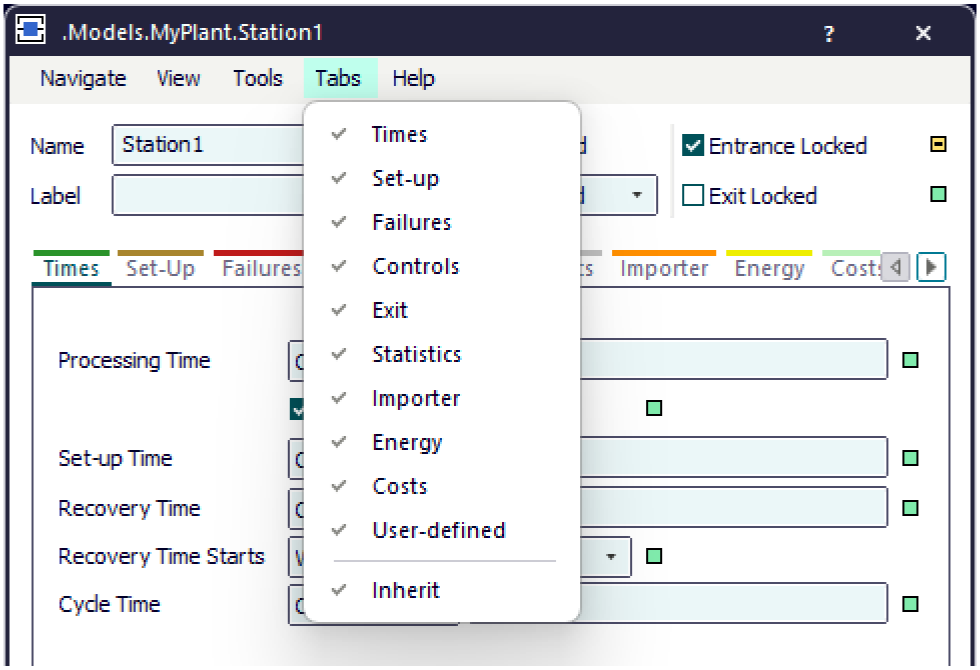
Refresh [on View menu]	Backward Blocking List
Show Statistics Report [on View menu]	Exit Blocking List
Show Attributes and Methods [on View menu]	Associated Lockout Zones
Contents [material flow objects]	Associated Shift Calendar
Forward Blocking List	

Tools Menu

The commands are described under the **Tools Menu**.

Tabs Menu

Use the commands of the **Tabs** menu to show or hide individual tabs of the selected material flow objects. If you hide tabs that you do not need, *Plant Simulation* opens the dialog faster, and you can change to those tabs faster that you need in your daily work.



Remarks

To apply the changed settings, click **OK**, close the dialog, and reopen it

The menu shows a check mark to the left of the displayed tabs.

The command **Inherit** turns inheritance of the displayed or hidden tabs in the dialog off or on.

Help Menu

The commands are described under the **Help Menu**.

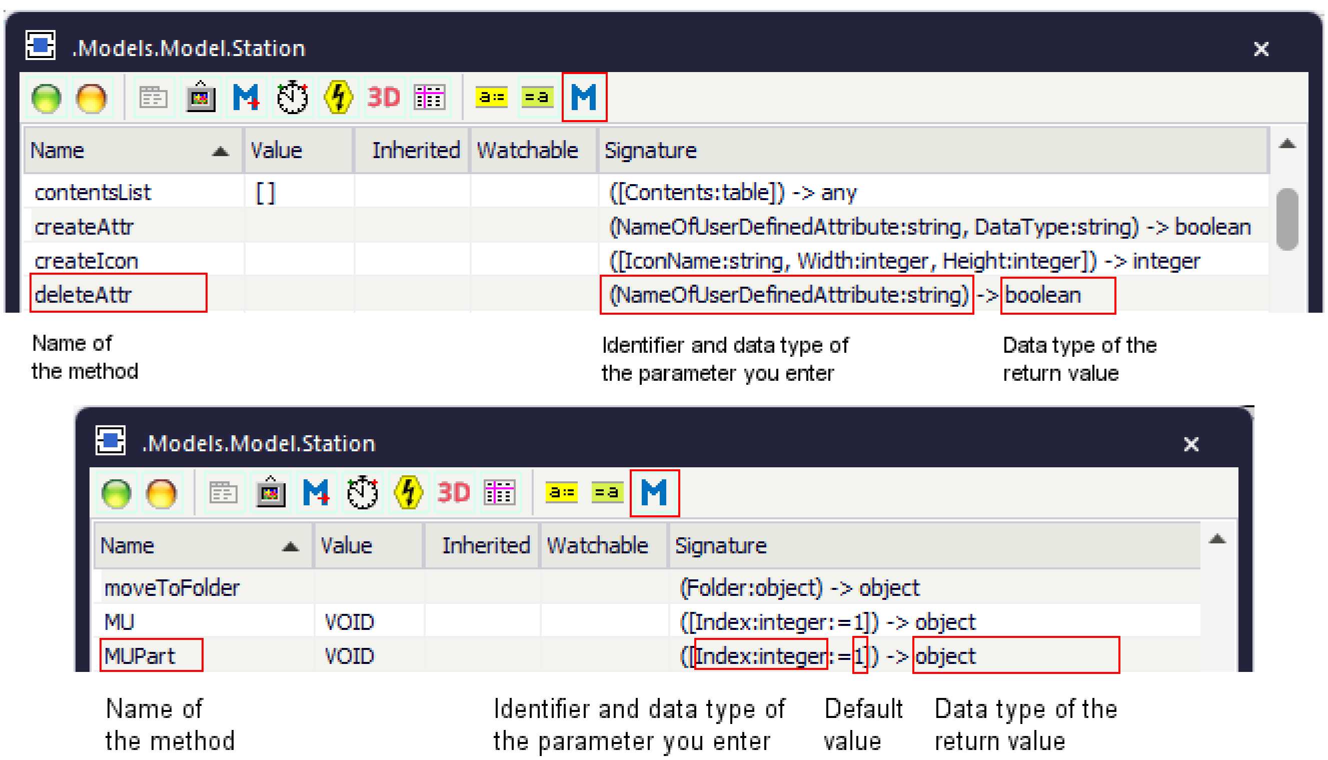
Methods of the Turnplate

_Methods of the Turnplate

The *Turnplate* provides:

- The *methods* listed in the table of contents to the left.
- The **_Methods of Curved Objects**.
- The **Methods of the Material Flow Objects**.
- The **Methods of All Objects**.

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window **Show Attributes and Methods**. The figure below illustrates the information using the example of the object *Station*.



- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected **Class [general description]**.